Monica Dessole

Curriculum Vitae

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Interests

I am particularly interested in high-performance scientific computing, parallel and distributed computing, open source software and new technologies. My work and research are primarily concerned with the design, validation and performance evaluation of novel algorithms and the implementation of high-quality scientific software.

Experience

Since 01.2026 **Research Software Engineer**, NATO Centre for Maritime Research and Experimentation (CMRE), Autonomous Naval Mine Warfare (ANMW) team
Project: Multi-GPU finite-difference (FD) solver for underwater acoustics.
- Implementation of a multi-GPU framework for a single-GPU nested grid finite difference 3-dimensional solver.

07.2023–12.2025 **Research Software Engineer**, CERN, EP-SFT group, ROOT team
Project: Heterogeneous computing data analytics frameworks for High Energy Physics.
- Design and development of algorithms for 4-vectors operations via CUDA and SYCL for heterogeneous hardware execution (multi-vendors GPUs, other accelerators like Risc-V, FPGAs)
- Maintenance and development of modern C++ codebase within an AGILE framework
- Supervision of junior fellows and BSc/MSc students

02.2022–06.2023 **Research Software Engineer**, Leonardo S.p.A., Research Labs, HPC/Cloud group
Project 1: Cloud application development based on microservice architectures through docker containerization for Natural Language Processing (NLP) applications.
- Python programming (PyTorch) and infrastructure components (Docker containers, VMs) for LLM training and fine-tuning
- Deployment on cloud IaaS (OpenStack)

Project 2: Signal processing for predictive maintenance
- Statistical analysis of time series aimed at identifying correlations with component breakage (NumPy, Pandas)

Project 3: Codebase porting of weather nowcasting software from IDL (Interactive Programming Language) to Python for "Protezione Civile"
- Codebase refactoring and modernization, Python programming

2017–18 **Research Fellow (Assegnista di Ricerca)**, Information Enginering Department, Università degli Studi di Padova
Project: Parallel sparse triangular approximate solvers on GPUs with application to ILU preconditioning for incompressible Navier-Stokes equations in the context of real-time fluid flow simulations for virtual prototyping.
- Design and development of low level linear algebra kernels
- Python and C/CUDA programming, code optimization and tuning

Education

2018–22 **Ph.D. in Computational Mathematics**, Università degli Studi di Padova
Thesis: "Topics in Numerical Linear Algebra for High-Performance Computing".
Project: GPU computing in C/CUDA applied to the parallel solution of problems arising from optimal control applications. HPC algorithms for rank-deficient problems with application to sparse recovery and compressed sensing problems.
Visiting: IRIT Toulouse, France, February 2020.

2017 **Master's Degree in Mathematics**, Università degli Studi di Padova
"ERASMUS+ Programme": Master 2 Calcul Scientifique, UFR de Mathématiques at Université Lille 1 - Sciences et Technologies, France, Sep. 2015 – Feb. 2016.

2014 **Bachelor's Degree in Mathematics**, Università degli Studi di Padova

Publications

[1] J. Chen, M. Dessole, A.L. Varbanescu, *"Component-Based Analytical Modeling of GPU Runtime Performance: a Case-Study in Scientific Computing"*. ICPE '25: Proceedings of the 16th ACM/SPEC International Conference on Performance Engineering, 2025.

[2] M. Dessole, J. Chen, A. Naumann *"GenVectorX: A performance-portable SYCL library for Lorentz Vectors operations"*. To appear on Journal Of Physics: Conference Series, 2024.

[3] J. Chen, M. Dessole, A.L. Varbanescu, *"Migrating CUDA to SYCL: A HEP Case Study with ROOT RDataFrame"*. IWOCL '24: Proceedings of the 12th International Workshop on OpenCL and SYCL, 2024.

[4] M. Dessole, F. Marcuzzi, *"Accurate detection of hidden material changes as fictitious heat sources from thermographic data"*. Numerical Heat Transfer, Part B: Fundamentals, 2023.

[5] M. Dessole, M. Dell'Orto, F. Marcuzzi, *"The Lawson-Hanson Algorithm with Deviation Maximization: Finite Convergence and Sparse Recovery"*. Numerical Linear Algebra with Applications, 2023.

[6] M. Dessole, F. Marcuzzi, *"Deviation Maximization for Rank Revealing QR factorizations"*. Numerical Algorithms, 2022.

[7] M. Dessole, F. Marcuzzi, M. Vianello *"dCATCH—A Numerical Package for d-Variate near G-Optimal Tchakaloff Regression via Fast NNLS"*. Mathematics, 2020.

[8] M. Dessole, F. Marcuzzi, *"A massively-parallel algorithm for Bordered Almost Block Diagonal systems on GPUs"*. Numerical Algorithms, 2020.

[9] M. Dessole, F. Marcuzzi, M. Vianello *"Accelerating the Lawson-Hanson NNLS solver for large-scale Tchakaloff regression designs"*. Dolomites Research Notes on Approximation, 2020.

[10] M. Dessole, F. Marcuzzi, *"Fully iterative ILU preconditioning of the unsteady Navier-Stokes equations for GPGPU"*. Computers & Mathematics with Applications, 2019.

Awards, personal funding and grants

2024 IWOCL '24 Best Poster Award

2022 Kovalevskaya grant for on-site participation at "ICM2022 – International Congress of Mathematicians" funded by Unione Matematica Italiana, Saint Petersburg, Russia (on-site event later cancelled)

2021 Participation grant for "Moxoff Academy" funded by Moxoff SpA, Milan, Italy

2021 Participation grant for "Model Order Reduction and Applications" funded by Fondazione CIME, Cetraro, Italy

2019 Participation grant for "Gene Golub SIAM Summer School on High Performance Data Analytics" funded by SIAM, Aussois, France

2018 PhD fellowship funded by beanTech Srl for three years doctoral studies at Università degli Studi di Padova, Italy

Conferences, seminars and schools

Invited presentations

16–12.06.2026 *"HPC driving AI" Summer School*, Heilbronn, Germany
Talk title: "ROOTing performance-portable heterogeneous computing with SYCL"

10–12.06.2024 *International Conference On Preconditioning Techniques For Scientific and Industrial Applications*, Georgia Institute of Technology Atlanta, USA
Talk title: "A fine-grained fully iterative ILU preconditioner for unsteady density variable Navier-Stokes equation"

24.02.2023 KIT, Karlsruhe, Germany
Seminar title: "GPU algorithms for the numerical solution of density dependent Navier Stokes equations"

17–28.06.2019 *Gene Golub SIAM Summer School (G2S3) on High Performance Data Analytics*, Aussois, France
Poster title: "GPGPU for the direct solution of BABD systems"

Contributed presentations

11–15.03.2024 "*International Workshop on Advanced Computing and Analysis Techniques in Physics Research*", Stony Brook, NY, USA
Talk title: "An empirical performance-portability evaluation for Lorentz Vectors computations via SYCL"

5–6.09.2022 "*Challenges in Numerical Analysis and Scientific Computing*", Braga, Portugal
Talk title: "A block pivoting strategy for fast RRQR"

23–27.05.2022 "*800 UniPD – 100 UMI*", Padova, Italy
Talk title: "Sparse recovery via fast nonnegative least squares"

14–15.02.2022 "*Due giorni di Algebra Lineare Numerica*", Naples, Italy
Talk title: "Deviation Maximization for rank-deficient problems"

28.05.2021 "*Rita PhD Seminar*", Online
Seminar title: "Numerical Linear Algebra for Caratheodory-Tchakaloff compression"

15–18.01.2020 "*Multivariate Approximation: Theory and Applications*", Perugia, Italy
Poster title: "Efficient computation of large-scale Tchakaloff regression designs"

11–12.07.2019 "*Sparse Days*", Toulouse, France
Talk title: "A massively-parallel algorithm for BABD systems on GPUs"

18–19.02.2019 "*Due giorni di Algebra Lineare Numerica*", Rome, Italy
Talk title: "Solving ABD systems on GPUs"

3–4.05.2018 "*Seminari Padovani di Analisi Numerica*", Padova, Italy
Talk title: "On the Approximate Solution of Sparse Triangular Systems on GPUs"

8–9.02.2018 "*Due giorni di Algebra Lineare Numerica e Applicazioni*", Padova, Italy
Talk title: "On the Approximate Solution of Sparse Triangular Systems for Massively Parallel Machines"

Organization

27–28.05.2026 "*International Conference on Preconditioning Techniques for Scientific and Industrial Applications (Precond26)*", Edinburgh, UK

2–6.03.2026 "*SIAM Conference on Parallel Processing for Scientific Computing (PP26)*", Berlin, Germany

Technical skills

- Proficient in C, CUDA, MPI, SYCL, C, Python
- Competent with C++, OpenMP, Matlab
- Good knowledge of Linux-based operating system
- Competent with profiling tools such as perf, valgrind, NVIDIA Nsight
- Comfortable with HPC job scheduling systems, e.g. PBS, Slurm
- Excellent knowledge of NLA libraries, e.g BLAS, LaPACK, MAGMA, cuBLAS, cuSPARSE
- Competent with Python data analysis frameworks, such as NumPy, Pandas, SciPy, Matplotlib
- Comfortable with Git version control system and agile software development (CI/DC)
- Competent with Docker, Virtual Machines deployment and cloud computing infrastructure management through OpenStack
- Familiar with SQL database administration

Languages

Italian (native), English (fluent), French (intermediate)